Rules of Simple Design,[4] and Gary's posts "Test Isolation Is About Avoiding Mocks" and "TDD, Straw Men, and Rhetoric."[5] [6]

Why Speed Is Important

Before I explain how you'll use this information to speed up your tests, let me make a pitch for the idea of being a little obsessive about test speed.

You're going to run your tests a lot.

If you're doing the kind of rapid-feedback TDD that I've been discussing up until now, you'll be running at least a subset of your tests nearly continuously, multiple times an hour. (Any exact number that I put here is going to seem shockingly high to people who aren't used to super-fast test feedback and shockingly low to those who are.) And you're probably not the only developer on your team. And it's likely that your entire test suite runs when you commit code to your repository via some kind of continuous build system.

The speed of a test run, whether it is your entire suite or a subset you're running in development, lands in one of seven groups:

1. So fast you don't even notice.
2. Fast enough for you to watch the test run without breaking focus.
3. Slow enough that you break focus but don't have time to do anything else.
4. Slow enough for you to get a cup of coffee.
5. Slow enough for you to eat lunch.
6. Slow enough for you to sleep.
7. Effectively infinite without a big parallelized server.

A project will have thousands of runs. Tens of thousands. Hundreds of thousands if the project is big. The difference between being in group 2 and group 3 becomes a big deal when you're trying to run the tests as often as you can. There are numerous studies that say that breaking focus is a huge drain in developer productivity.

The difference between a 10-second run in group 2 and a 30-second run in group 3 is important. A group 2 run allows you get feedback without breaking focus. A group 3 run means you lose your train of thought. If you've ever been

4. https://leanpub.com/4rulesofsimpledesign
5. https://www.destroyallsoftware.com/blog/2014/test-isolation-is-about-avoiding-mocks
6. https://www.destroyallsoftware.com/blog/2014/tdd-straw-men-and-rhetoric